Melanoma–Pancreatic Cancer Syndrome (MONDO:0011713, OMIM 606719): A Comprehensive Disease Characterization

> **Causal gene: `CDKN2A` (9p21.3), encoding the dual tumor suppressors p16INK4a and p14ARF.** This report characterizes the autosomal dominant hereditary cancer-predisposition syndrome catalogued as OMIM 606719 / MONDO:0011713, in which germline loss-of-function variants in *CDKN2A* predispose to cutaneous melanoma and pancreatic ductal adenocarcinoma (PDAC). It is a specific presentation within the Familial Atypical Multiple Mole Melanoma–Pancreatic Cancer (FAMMM-PC) spectrum. Although the name references melanoma and pancreatic cancer, and the disease is sometimes conflated with FAMMM (OMIM #155601) or CDK4-driven melanoma (OMIM #609048), the molecular lesion of this specific entity is in *CDKN2A* — not CDK4, BRCA2, or PALB2 — and all identifiers above point to the *CDKN2A*-driven entity.

MONDO: MONDO:0011713 · **OMIM:** 606719 · **Causal gene:** CDKN2A (9p21.3) · **Category:** Mendelian, autosomal dominant · **Report date:** 2026-11-05

Summary

Melanoma–Pancreatic Cancer Syndrome is an autosomal dominant hereditary cancer predisposition syndrome caused by germline loss-of-function variants in **CDKN2A** on chromosome 9p21.3. This locus is unusual in that it encodes **two distinct tumor suppressor proteins from overlapping reading frames**: p16INK4a (which inhibits CDK4/CDK6 to maintain RB-mediated G1 cell-cycle arrest) and p14ARF (which stabilizes p53 by inhibiting MDM2). Germline inactivation of CDKN2A therefore removes a "pre-installed" driver lesion that both melanocytes and pancreatic ductal epithelium would otherwise need to acquire somatically, lowering the threshold for malignant transformation. The clinical hallmark is the

co-occurrence, within families, of **cutaneous melanoma** (often early-onset, multiple primaries, with clinically atypical nevi) and **pancreatic ductal adenocarcinoma**, frequently accompanied by a broader spectrum of tobacco- and UV-potentiated cancers.

Penetrance is high but strongly **ascertainment- and environment-dependent**. In the Dutch founder cohort (p16-Leiden, c.225_243del), carriers face roughly a **70% lifetime melanoma risk and a 15–20% lifetime pancreatic cancer risk**. Population-based estimates of melanoma penetrance are considerably lower (~28% by age 80) than those derived from dense multiple-case kindreds (58–91%), and penetrance rises with cumulative UV exposure across geographic latitude. Tobacco smoking dramatically amplifies the risk of pancreatic, respiratory, and upper-digestive cancers, a clear gene–environment interaction, while MC1R variants act as cosegregating genetic modifiers of melanoma risk.

Management is **cancer-specific rather than syndrome-specific**: there is no therapy for the germline defect itself. Melanoma is managed by surveillance (total-body photography, dermoscopy, sequential digital dermoscopic imaging), wide local excision, and — for advanced disease — modern immune checkpoint inhibitors and BRAF/MEK-targeted therapy. For pancreatic cancer, surgical resection of screen-detected early-stage tumors is the only route to cure; annual MRI/MRCP ± endoscopic ultrasound surveillance of carriers (endorsed by the International Cancer of the Pancreas Screening [CAPS] Consortium) improves resectability, long-term cure, and is cost-effective. Smoking cessation and sun protection are the cornerstones of primary prevention, and genetic counseling with cascade testing is central to family management.

Key Findings

Finding 1 — CDKN2A germline loss-of-function is the cause (F001)

Germline pathogenic variants in **CDKN2A** (9p21), encoding two tumor suppressors (**p16INK4a** and **p14ARF**) from alternative reading frames, cause this autosomal dominant syndrome of melanoma plus pancreatic ductal adenocarcinoma. Approximately **30–40% of dense FAMMM/**

PCMS families carry detectable CDKN2A mutations. p16INK4a inhibits CDK4/CDK6 to maintain RB-mediated G1 arrest; p14ARF stabilizes p53 via MDM2 inhibition — so a single locus guards both of the two canonical tumor-suppressor axes (RB and p53).

- PMID: 32941720: *"About 30% of PCMS cases carry mutations in CDKN2A. This gene encodes several protein isoforms, one of which, known as p16, regulates the cell-cycle by interacting with CDK4/CDK6 kinases."*
- PMID: 14871223: *"germline mutations in the INK4a antioncogene encoding p16 protein were found on average in approximately 40% of the FAMMM syndrome."*

Finding 2 — p16-Leiden penetrance and the value of pancreatic surveillance (F002)

The Dutch founder mutation **c.225_243del** (a 19-bp deletion in exon 2, "**p16-Leiden**") confers roughly a **70% lifetime melanoma risk** and a **15–20% lifetime pancreatic cancer risk**. In a surveillance cohort of 347 carriers undergoing annual MRI ± EUS, PDAC was detected in 8.9%, was resectable in 71%, and long-term cure among resected patients was estimated at **47.1%**. Modeling showed surveillance reduced PDAC mortality by 12.1% and added 2.10 years of life expectancy at ~€14,000/QALY — i.e., cost-effective.

- PMID: 31203567: *"Carriers of pathogenic variants in CDKN2A have a 70% life-time risk of developing melanoma and 15-20% risk of developing pancreatic cancer (PC)."*
- PMID: 36785917: *"PDAC was detected in 31 patients (8.9%) and the tumor could be resected in 22 patients (71.0%). Long-term cure among patients with resected PDAC was estimated at 47.1%."*

Finding 3 — Mouse models confirm the mechanism (F003)

In vivo genetics strongly supports the human mechanism. Ink4a-deficient mice are cancer-prone; melanocyte-specific activated **H-rasG12V on an ink4a-null background** produces spontaneous cutaneous melanoma with short latency and high penetrance, with loss of heterozygosity (LOH) at the wild-type ink4a allele. Tyr::NRAS(Q61K);Cdkn2a-/- mice develop metastasizing melanoma. Knock-in **Cdk4R24C** mice (the INK4-insensitive CDK4 allele that mirrors the human hereditary-melanoma CDK4 mutation) develop tumors with near-complete penetrance. Both the Ink4a/pRB and Arf/p53 arms contribute.

- PMID: 9353252: *"mice with melanocyte-specific expression of activated H-rasG12V on an ink4a-deficient background develop spontaneous cutaneous melanomas after a short latency and with high penetrance."*

- PMID: 11726500: *"This mutation (replacement of Arg24 by Cys)... renders Cdk4 insensitive to INK4 inhibitors... homozygous Cdk4(R24C/R24C) mutant mice develop multiple tumors with almost complete penetrance."* (Note: CDK4 R24C is the alternative hereditary-melanoma gene, which phenocopies p16 loss.)

Finding 4 — Broad extra-melanoma cancer spectrum and gene–environment interaction with tobacco (F004)

In a prospective Swedish cohort of p.Arg112dup (p16-113insArg) carriers, the relative risk versus controls was **43.8** (95% CI 13.8–139.0) for pancreatic cancer, **17.1** (6.3–46.5) for upper-digestive-tract cancer, **15.6** (5.4–46.0) for respiratory cancer, and **5.0** (3.7–7.3) for non-melanoma cancers overall. Critically, **ever-smoking carriers** versus never-smoking carriers had an odds ratio of **9.3** (95% CI 1.9–44.7) for pancreatic/respiratory/upper-digestive cancers — a clear gene–environment interaction. Head-and-neck/oral SCC is also over-represented. MC1R variants cosegregate as melanoma modifiers, and UV exposure modulates penetrance across latitude.

- PMID: 24935963: *"In carriers prospective RR for non-melanoma cancers was 5.0 (95% CI 3.7 to 7.3), for pancreatic cancer 43.8 (95% CI 13.8 to 139.0), for cancers in upper digestive tissues 17.1 (95% CI 6.3 to 46.5), and in respiratory tissues 15.6 (5.4 to 46.0)."*
- PMID: 24935963: *"In ever-smoking carriers compared with never-smoking carriers, the odds ratio (OR) of cancers in pancreas, respiratory or upper digestive tissues was 9.3 (95% CI 1.9 to 44.7)."*
- PMID: 16462187: *"Certain variants in a low-penetrance gene, MC1R... act as a genetic modifier when cosegregating with a deleterious p16 gene. The penetrance of these melanoma-predisposing genes is largely influenced by ultraviolet exposure across geographic latitude."*

Finding 5 — Cutaneous phenotype: atypical nevi, early-onset and multiple primary melanomas (F005)

CDKN2A-positive patients develop melanoma **earlier** and are prone to **multiple primary melanomas**. In Swedish carriers, melanoma occurred only in mutation carriers (median age 36 years) and correlated with clinically atypical nevi (CAN, P=0.007) and red hair/MC1R, but

carriers did **not** uniformly meet full atypical-mole-syndrome criteria (median nevus count 24). Atypical mole syndrome raises melanoma risk but cannot reliably predict germline mutation status.

- PMID: 20526219: *"A positive mutation status (CDKN2A) was correlated to one or more CAN (P=0.007) but neither to blue eyes, red hair colour, heavy freckling nor high number of nevi."*
- PMID: 16462187: *"Familial melanoma patients develop melanomas earlier and are prone to developing multiple primary melanomas."*

Suggested HPO terms: Cutaneous melanoma (HP:0012056), Dysplastic nevus (HP:0003764), Pancreatic neoplasm (HP:0002894), Neoplasm of the skin (HP:0008069).

Finding 6 — Treatment is tumor-specific (F006)

There is no therapy for the germline syndrome itself; management targets individual tumors. **Melanoma:** wide local excision for localized disease; for advanced/resectable stage III–IV, immune checkpoint inhibitors (anti-PD-1 nivolumab/pembrolizumab, anti-CTLA-4 ipilimumab, anti-LAG-3 relatlimab) and BRAF/MEK inhibitors (for BRAF-V600 tumors). Neoadjuvant ipilimumab + nivolumab improved event-free survival versus adjuvant nivolumab in the NADINA trial. **Pancreatic cancer:** surgical resection of screen-detected early PDAC is the only route to cure (~47% long-term cure for resected screen-detected tumors). CDKN2A loss produces CDK4/6 dependence, providing a mechanistic rationale for CDK4/6 inhibitors (investigational in CDKN2A-null tumors).

- PMID: 42794996: *"In resectable macroscopic stage III melanoma, neoadjuvant ipilimumab plus nivolumab with response-driven postoperative management improved event-free survival versus adjuvant nivolumab in NADINA."*

Suggested NCIT terms: Pembrolizumab (C106432), Nivolumab (C68924), Ipilimumab (C2654), Dabrafenib (C82386), Trametinib (C77908), Pancreaticoduodenectomy (C51782).

Finding 7 — Molecular mechanism: p16 destabilization / loss of CDK4 binding → RB hyperphosphorylation → unchecked G1/S (F007)

Molecular dynamics/docking of melanoma-prone p16 mutants shows that many mutations (e.g., G23D, P114S, A60V) **abolish CDK4 binding** while others (R24Q, G67R) partially impair it, disrupting a positive electrostatic patch on the binding surface and lowering complex stability

— often without gross structural change. Tumor-derived mutations (P81L, V126D) **destabilize the N-terminal half** of p16, which contains CDK-binding residues. The CDK4 R24C mutation renders CDK4 insensitive to INK4 inhibition, phenocopying p16 loss.

- PMID: 39821174: *"Despite no significant structural changes in p16 due to mutation, the binding affinity was found to be affected... specific mutations, such as G23D, P114S, and A60V resulted in loss of binding to CDK4."*
- PMID: 12009890: *"the N-terminal half of p16 is significantly more sensitive to proteolysis in both tumor-derived mutant proteins than in the wild type, suggesting that this region is particularly unstable."*

Finding 8 — CAPS Consortium endorses pancreatic surveillance for CDKN2A carriers (F008)

The International Cancer of the Pancreas Screening (CAPS) Consortium formally lists **p16 (CDKN2A) carriers with ≥1 affected first-degree relative** as candidates for pancreatic surveillance. The goal is to detect and treat **T1N0M0** margin-negative PDAC and high-grade dysplastic precursors (PanIN/IPMN) using **EUS and/or MRI/MRCP** rather than CT/ERCP. The 2020 update reached consensus on 55 statements; surveillance should start no earlier than age 50 (or 10 years before the youngest affected relative), with germline-mutation carriers potentially starting earlier, in expert multidisciplinary centers. CAPS cyst-management criteria are moderately specific (85%) but insufficiently sensitive (60%) for advanced neoplasia.

- PMID: 23135763: *"the following were candidates for screening: first-degree relatives (FDRs) of patients with PC from a familial PC kindred with at least two affected FDRs; patients with Peutz-Jeghers syndrome; and p16, BRCA2 and hereditary non-polyposis colorectal cancer (HNPCC) mutation carriers with ≥ 1 affected FDR."*
- PMID: 31672839: *"surveillance should start no earlier than age 50 or 10 years earlier than the youngest relative with pancreatic cancer."*

Finding 9 — Melanoma penetrance is ascertainment-dependent (F009)

In a **population-based** kin-cohort sample, melanoma risk in CDKN2A carriers was **14%** (95% CI 8–22) by age 50, **24%** (15–34) by age 70, and **28%** (18–40) by age 80 — markedly lower than estimates from high-density multiple-case families (**58% in Europe to 91% in Australia** by age 80). This gradient reflects both ascertainment bias and cumulative UV exposure by latitude.

- PMID: 16234564: *"The risk of melanoma in CDKN2A mutation carriers was approximately 14% (95% CI = 8% to 22%) by age 50 years, 24% (95% CI = 15% to 34%) by age 70 years, and 28% (95% CI = 18% to 40%) by age 80 years."*
- PMID: 16234564: *"the lifetime risk... of melanoma in CDKN2A mutation carriers is very high, ranging from 58% in Europe to 91% in Australia by age 80 years."*

Finding 10 — CDKN2A is a core somatic PDAC driver, validating the germline mechanism (F010)

Deep sequencing of 499 PDAC patients identified recurrent somatic driver mutations in **KRAS, TP53, CDKN2A, SMAD4, ARID1A, RNF43**; ~90% of PDAC is driven by (currently untargetable) KRAS mutations. The same KRAS-activating + CDKN2A/TP53/SMAD4-inactivating quartet drives PDAC and its variants. This mirrors the germline-plus-second-hit logic of the syndrome: germline CDKN2A loss supplies one of the obligatory PDAC driver lesions, with somatic KRAS activation and TP53/SMAD4 loss completing transformation.

- PMID: 36999792: *"revealed somatic driver mutations in KRAS, TP53, CDKN2A, SMAD4, ARID1A, RNF43."*
- PMID: 28722124: *"inactivating mutations in the tumor suppressor genes CDKN2A, TP53, and SMAD4."*

Finding 11 — Epidemiology: rare Mendelian subset of familial melanoma (F011)

Approximately **5–10% of cutaneous melanoma** occurs in individuals with a family history. CDKN2A germline mutations occur in ~5–20% of familial melanoma and ~30–40% of dense FAMMM/PCMS kindreds, but are **rare in unselected/sporadic** melanoma (e.g., none among 125 consecutive Latvian cases). The Dutch national database catalogued 172 families with 15

distinct pathogenic CDKN2A variants (649 carriers), the founder c.225_243del (p16-Leiden) being most prevalent. The syndrome is rare (Orphanet FAMMM ORPHA:618). Both sexes are affected.

- PMID: 40851494: *"Approximately 5-10% of cutaneous melanoma occurs in individuals with a family history of the disease. While known high-penetrance genes, such as CDKN2A, explain some cases."*
- PMID: 32482799: *"We identified 172 families in which 649 individuals carried 15 different pathogenic variants. The most prevalent variant was the founder mutation c.225_243del."*

Comprehensive Section-by-Section Characterization

1. Disease Information

Overview. Melanoma–Pancreatic Cancer Syndrome (also widely described within the Familial Atypical Multiple Mole Melanoma–Pancreatic Cancer, FAMMM-PC, spectrum) is an autosomal dominant hereditary cancer predisposition syndrome. Affected families show clustering of cutaneous malignant melanoma and pancreatic ductal adenocarcinoma, frequently with multiple atypical (dysplastic) nevi and additional tobacco/UV-related cancers.

Key identifiers. | Resource | Identifier | |---|---| | OMIM | 606719 | | MONDO | MONDO:0011713 | | Orphanet | ORPHA:618 (FAMMM) | | Causal gene | CDKN2A (HGNC:1787), 9p21.3 | | Related gene | CDK4 (alternative, rarer hereditary melanoma locus) |

Synonyms / alternative names: FAMMM-PC syndrome; Pancreatic cancer–melanoma syndrome (PCMS); melanoma–pancreatic cancer syndrome; CDKN2A-associated FAMMM. Related OMIM entries: cutaneous malignant melanoma 2 (CMM2, OMIM 155601); FAMMM (OMIM 155601); CDK4-related melanoma (OMIM 609048).

Source of information: Primarily aggregated disease-level resources (OMIM, Orphanet) and family-/cohort-based studies (Dutch, Swedish, Italian, Australian/North American registries), plus individual case reports.

2. Etiology

Causal factor: Germline loss-of-function variants in **CDKN2A** (p16INK4a and/or p14ARF). This is the primary, necessary genetic cause (F001). A minority of hereditary melanoma families instead carry **CDK4** gain-of-function variants (R24C/R24H) that phenocopy p16 loss (F003).

Genetic risk factors: Pathogenic CDKN2A variants (e.g., p16-Leiden c.225_243del; p.Arg112dup; p.Arg24Gln; splice and ARF-specific variants). **Modifier:** MC1R variants raise melanoma risk and cosegregate with deleterious p16 (F004). A TERT/CLPTM1L SNP was investigated as a candidate modifier ( ).

P 31203567

Environmental risk factors: UV radiation (melanocytes) and **tobacco smoking** (pancreas, respiratory and upper-digestive epithelium) are the dominant modifiable exposures. Age, sex (both affected), and family history further stratify risk. General PDAC risk factors (diabetes, obesity, alcohol) also apply ( ).

P 37417291

Protective factors: Sun protection and smoking avoidance reduce penetrance (inferred from the inverse of the documented risk factors). No validated germline protective allele is established for this syndrome.

Gene–environment interactions: Strongly demonstrated — ever-smoking carriers have ~9-fold higher risk of pancreatic/respiratory/upper-digestive cancers than never-smoking carriers (OR 9.3), and UV exposure across latitude modulates melanoma penetrance (F004).

3. Phenotypes

Phenotype	Type	HPO	Onset	Frequency / notes
Cutaneous melanoma	Neoplasm	HP:0012056	Early-adult (median ~36 y in carriers)	High; often multiple primaries
Pancreatic ductal adenocarcinoma	Neoplasm	HP:0002894	Adult (mean ~60 y)	15–20% lifetime (p16-Leiden)
Dysplastic / clinically atypical nevi	Physical sign	HP:0003764	Childhood–adult	Common but not obligate
Multiple primary melanomas	Neoplasm	HP:0012056	Adult	Characteristic
Head-and-neck / oral SCC	Neoplasm	—	Adult	Over-represented (P 35123577)
Respiratory / upper-digestive cancers	Neoplasm	—	Adult	Elevated, tobacco-potentiated

Severity is high (life-threatening malignancies); progression is tumor-dependent. Quality-of-life impact arises from lifelong surveillance burden, repeated excisions, and the psychological weight of cancer risk; PDAC carries severe QoL and mortality impact. Carriers do **not** uniformly meet full atypical-mole-syndrome criteria (F005).

4. Genetic / Molecular Information

Causal gene: CDKN2A (HGNC:1787; OMIM 600160), 9p21.3. The locus encodes p16INK4a (transcript using exons 1α-2-3) and p14ARF (exon 1β + alternative reading frame of exon 2), two unrelated proteins (F001).

Pathogenic variants: Missense (e.g., p.Arg24Gln, p.Gly23Asp, p.Ala60Val, p.Pro114Ser), in-frame deletions (c.225_243del / p16-Leiden), duplications (p.Arg112dup), splice-site, and ARF-specific lesions. ClinVar/ACMG classifications range pathogenic → likely pathogenic → VUS (e.g., Ala148Thr is a low-penetrance polymorphism, P 17505264).

Allele frequencies are very low in gnomAD (rare founder alleles region-specific). **Origin:**

germline for the syndrome; CDKN2A is also a recurrent **somatic** PDAC/melanoma driver (F010). **Functional consequence:** loss of function (destabilization, loss of CDK4 binding) (F007).

Modifier genes: MC1R (melanoma penetrance), candidate TERT/CLPTM1L SNP. **Chromosomal:** second-hit LOH at 9p21 in tumors (F003). **Epigenetic:** CDKN2A promoter hypermethylation is a well-known somatic silencing mechanism in sporadic tumors and can contribute to the "second hit" in carriers.

5. Environmental Information

Environmental factors: UV radiation (principal melanoma driver). **Lifestyle factors:** tobacco smoking (strong GxE amplifier of pancreatic/aerodigestive risk); alcohol and obesity/diabetes are general PDAC risk factors (P 37417291).

Infectious agents: none implicated.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **germline CDKN2A loss-of-function variant** is inherited (present in every cell). *(leads to)*
2. **Reduced or absent functional p16INK4a** (and, for many variants, p14ARF), due to protein destabilization and/or loss of CDK4 binding — demonstrated biophysically (F007). *(results in)*
3. **De-repression of CDK4/6–cyclin D kinase activity** (p16 normally inhibits CDK4/6). *(leads to)*
4. **Hyperphosphorylation and inactivation of RB.** *(results in)*
5. **E2F-driven transcription and unchecked G1→S transition**, i.e., escape from senescence/cell-cycle arrest. *(enables)*
6. Acquisition of a **somatic second hit** inactivating the wild-type CDKN2A allele (LOH) — observed in tumors (F003, inferred from mouse LOH). *(combines with)*
7. **Carcinogen-driven oncogenic RAS/MAPK activation** — UV-induced in melanocytes, tobacco-associated KRAS activation in pancreatic ductal cells (F003, F010). *(drives)*
8. **Malignant transformation** of the melanocyte (→ melanoma) or pancreatic ductal epithelial cell (→ PDAC, typically alongside somatic TP53/SMAD4 loss). *(manifests as)*

9. **Clinical melanoma and/or pancreatic ductal adenocarcinoma**, and a broader tobacco/UV-potentiated cancer spectrum.

Branch (p14ARF arm): For variants that also disable p14ARF, loss of ARF → failure to inhibit MDM2 → **degradation of p53** → impaired apoptosis/senescence, adding a second oncogenic axis (F001,

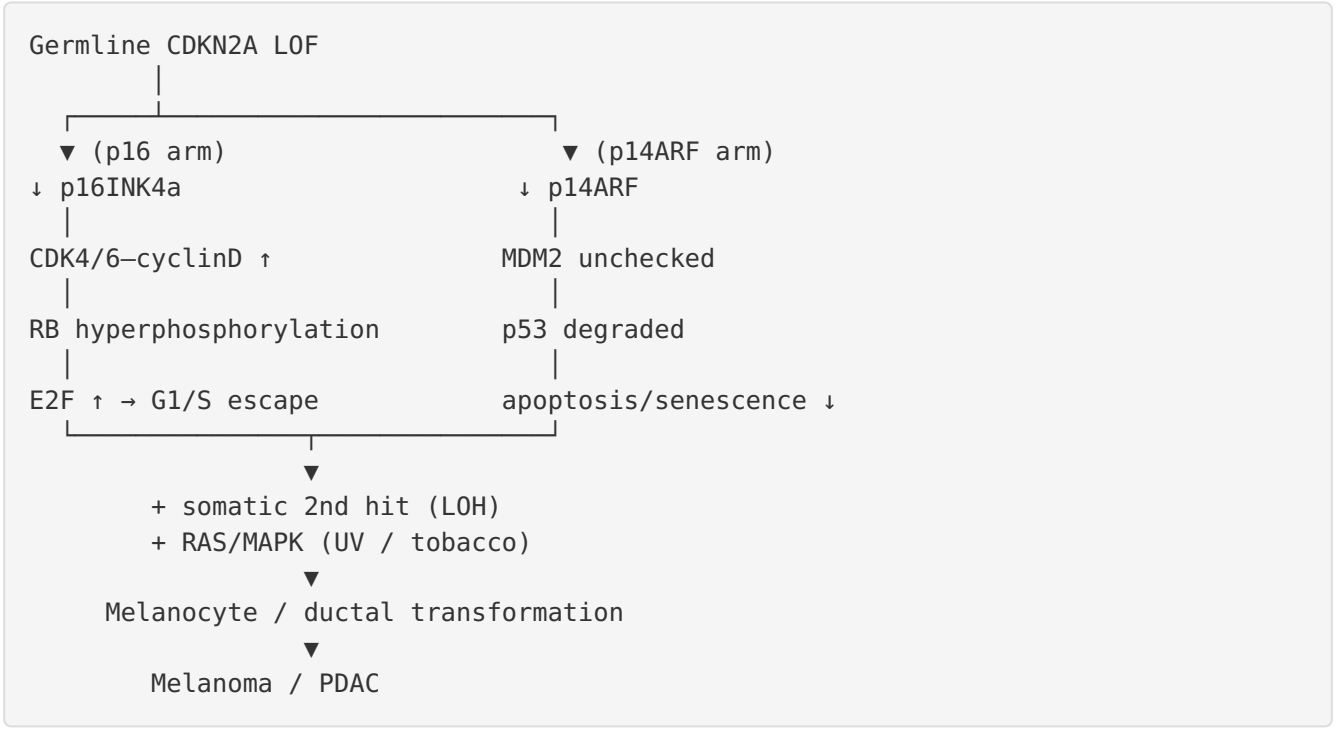
P 12406321

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Thus a single locus can cripple both the RB and p53 tumor-suppressor pathways. The independent contribution of the ARF arm in humans is partly inferred rather than fully demonstrated.



Molecular pathways: RB/cell-cycle (CDK4/6–cyclin D–RB–E2F), p53/MDM2, RAS–MAPK. **Cellular processes:** cell-cycle dysregulation, senescence escape, impaired apoptosis. **GO terms:** cell cycle arrest (GO:0007050), regulation of G1/S transition (GO:2000045), cyclin-dependent protein kinase activity (GO:0004693), negative regulation of cell population proliferation (GO:0008285). **CL terms:** melanocyte (CL:0000148), pancreatic ductal cell (CL:0002079).

7. Anatomical Structures Affected

- **Primary organs:** skin (UBERON:0002097) — melanocytes; pancreas (UBERON:0001264) — exocrine ductal epithelium.
- **Secondary / extended spectrum:** respiratory tract (UBERON:0000065), upper digestive tract/oral cavity (UBERON:0000167), and other sites in heavily exposed carriers.
- **Tissue/cell level:** epithelial (pancreatic ductal cell, CL:0002079) and neural-crest-derived melanocyte (CL:0000148).
- **Subcellular:** nucleus (GO:0005634) — site of RB/E2F/p53 regulation; cytoplasmic p16–CDK4 interaction.
- **Lateralization:** not applicable (neoplastic, site-variable).

8. Temporal Development

Onset: Melanoma typically early adulthood (median ~36 y in carriers); PDAC typically later adulthood (mean ~60 y). Course is chronic lifelong cancer risk; individual tumors progress per standard oncologic staging (AJCC). PDAC is often rapidly progressive once invasive — surveillance aims to catch the pre-invasive/early-invasive window (T1N0M0), the critical period for curative intervention (F002, F008). Main-pancreatic-duct dilatation can be an early surveillance signal (P 30910317).

9. Inheritance and Population

- **Inheritance:** autosomal dominant. **Penetrance:** incomplete, age-dependent, ascertainment- and UV-dependent (melanoma ~28% by 80 population-based vs 58–91% in dense kindreds; PDAC 15–20% in p16-Leiden) (F002, F009).
- **Expressivity:** highly variable (melanoma-only, PDAC-only, or combined; variable extra-tumor spectrum).
- **Founder effects:** p16-Leiden (Netherlands); p.Arg112dup (Sweden); population-specific spectra (F011).
- **Epidemiology:** rare Mendelian subset; familial melanoma ~5–10% of all melanoma, CDKN2A carriers rare in unselected patients; both sexes affected (F011).
- **Genetic anticipation / mosaicism / consanguinity:** not characteristic (dominant, not a repeat-expansion disorder).

10. Diagnostics

- **Genetic testing:** single-gene CDKN2A sequencing/del-dup; multigene hereditary-melanoma/pancreatic-cancer panels (CDKN2A, CDK4, BAP1, POT1, BRCA2, PALB2, ATM); WES/WGS in undiagnosed families. GeneReviews/GTR-aligned.
- **Melanoma diagnostics:** dermoscopy, total-body photography, sequential digital dermoscopic imaging; histopathology/IHC; FISH for ambiguous melanocytic lesions (P 21834951).
- **Pancreatic diagnostics:** MRI/MRCP and EUS (preferred over CT/ERCP per CAPS); CA 19-9 as adjunct biomarker; liquid biopsy/ctDNA emerging (P 41671451).
- **Differential diagnosis:** other hereditary melanoma/pancreatic syndromes (BAP1-TPDS, Peutz-Jeghers, hereditary breast-ovarian [BRCA2/PALB2], Lynch).

11. Outcome / Prognosis

Prognosis is tumor- and stage-dependent. **Melanoma:** excellent for early/in-situ disease; modern adjuvant/neoadjuvant immunotherapy improves outcomes in resectable stage III (F006). **PDAC:** historically dismal (~5-year survival ~5% in general population), but **surveillance-detected, resected PDAC in carriers has ~47% long-term cure** (F002). Multiple/synchronous PDAC occurs, sometimes motivating total pancreatectomy (P 29769629).

Prognostic factors: stage at detection, resectability, smoking status. Pancreatic surveillance adds ~2.1 years life expectancy per cost-effectiveness modeling (F002).

12. Treatment

- **Melanoma pharmacotherapy:** anti-PD-1 (pembrolizumab NCIT:C106432, nivolumab NCIT:C68924), anti-CTLA-4 (ipilimumab NCIT:C2654), anti-LAG-3 (relatlimab); BRAF/MEK inhibitors (dabrafenib NCIT:C82386 + trametinib NCIT:C77908) for BRAF-V600 tumors. Neoadjuvant ipilimumab+nivolumab (NADINA) superior to adjuvant nivolumab (F006).
- **Surgical:** wide local excision (melanoma); pancreaticoduodenectomy/partial or total pancreatectomy (NCIT:C51782) for resectable PDAC.
- **Mechanism-based / investigational:** CDK4/6 inhibitors (palbociclib/ribociclib/abemaciclib) rational in CDKN2A-null tumors (F006, F007); PI3K/mTOR combinations explored in

CDKN2A-deficient contexts
(P 28704762).

- **Pharmacogenomics / personalized:** BRAF genotype guides targeted therapy; germline status guides surveillance and family testing.

13. Prevention

- **Primary:** sun protection/UV avoidance; **smoking cessation** (especially impactful given GxE, F004).
- **Secondary (early detection):** skin surveillance with TBP/SDDI
(P 24964862);
annual pancreatic MRI ± EUS in carriers per CAPS (F008); cascade genetic testing of relatives.
- **Genetic counseling:** risk assessment, family planning, cascade testing; behavioral-change counseling
(P 35372037).
- **Tertiary:** intensive post-resection surveillance for metachronous tumors.

14. Other Species / Natural Disease

- **Orthologous gene:** mouse *Cdkn2a* (Ink4a/Arf). **Taxonomy:** *Mus musculus* (NCBI:txid10090) extensively used.
- **Natural disease:** CDKN2A/p16 pathway alterations occur across many species' cancers; no single well-defined companion-animal "melanoma–pancreatic syndrome" ortholog is established in OMIA, but the pathway is deeply evolutionarily conserved.
- **Comparative biology:** mouse models faithfully reproduce the Ink4a-loss + RAS/MAPK melanoma mechanism, including LOH of the wild-type allele (F003).

15. Model Organisms

- **Mammalian (mouse):** Ink4a/Arf knockout; melanocyte-specific H-rasG12V;ink4a-null (spontaneous melanoma); Tyr::NRAS(Q61K);Cdkn2a-/- (metastasizing melanoma); Cdk4R24C knock-in (INK4-insensitive CDK4) (F003).
- **Phenotype recapitulation:** strong for melanoma predisposition, UV/RAS cooperation, and second-hit LOH; both Ink4a/pRB and Arf/p53 arms modeled.

- **Limitations:** mouse skin differs from human; spontaneous PDAC in these specific melanoma models is less emphasized than melanoma (pancreatic modeling typically uses KRAS-driven GEMMs).
- **Resources:** MGI; genetically engineered mouse lines; patient-derived cell lines for p16 structure–function
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  P 12009890
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  P 39821174

Mechanistic Model / Interpretation

The unifying interpretation is that **germline CDKN2A loss functions as a "pre-installed" driver hit**. Both melanoma and PDAC are multi-hit cancers; CDKN2A inactivation is a recurrent **somatic** driver in both tumor types (F010). A germline carrier is therefore born one obligatory step into the tumorigenic cascade in every cell, dramatically lowering the probability threshold for transformation once tissue-specific carcinogens supply the complementary lesions (UV→melanocyte; tobacco→pancreatic/aerodigestive epithelium). This single model explains:

- **Why both melanoma and PDAC cluster** in the same families (same germline driver, two tissues with potent carcinogen exposures).
- **Why penetrance tracks UV latitude and tobacco** (the complementary somatic hits are environmentally supplied) — the gene–environment interaction (F004) and the population-vs-kindred penetrance gap (F009).
- **Why mouse genetics reproduce the disease** only when Ink4a loss is combined with RAS/MAPK activation and the wild-type allele is lost (F003).
- **Why management is tumor-specific** and surveillance-centric: you cannot replace the germline tumor suppressor, so you intercept tumors early (F002, F006, F008).

Tissue	Pre-installed hit (germline)	Complementary somatic hit(s)	Environmental driver
Melanocyte	CDKN2A LOF	2nd-allele LOH; BRAF/NRAS	UV
Pancreatic duct	CDKN2A LOF	KRAS; TP53; SMAD4	Tobacco

Evidence Base

PMID	Role	Supports
32941720	Case/review	CDKN2A causality, p16–CDK4/6, 30% frequency (F001)
14871223	Review	FAMMM pathogenesis, ~40% mutation detection (F001)
31203567	Cohort	70% melanoma / 15–20% PDAC lifetime risk (F002)
36785917	Cohort/modeling	Surveillance resection/cure, cost-effectiveness (F002)
9353252	Mouse	Ink4a loss + Ras → melanoma (F003)
11726500	Mouse	CDK4 R24C knock-in tumors (F003)
26587692	Mouse	NRAS;Cdkn2a-/- metastasizing melanoma (F003)
12406321	Review	Ink4a/pRB and Arf/p53 arms in melanoma (F003/F007)
24935963	Prospective cohort	Extra-melanoma RR; tobacco GxE (F004)
16462187	Review	MC1R modifier, UV penetrance, early/multiple melanoma (F004/F005)
20526219	Cohort	CAN correlation, not full AMS (F005)
42794996	Review	Neoadjuvant immunotherapy (NADINA) (F006)
39821174	Computational	p16 mutations abolish CDK4 binding (F007)
12009890	In vitro	p16 N-terminal destabilization (F007)
15899796	In vitro/mouse	CDK4R24C, E2F/topoII mechanism (F007)
23135763	Guideline	CAPS: p16 carriers as screening candidates (F008)
31672839	Guideline	CAPS 2020: start age, modality (F008)
33593706	Guideline eval	CAPS cyst criteria performance (F008)
16234564	Population study	Ascertainment-dependent penetrance (F009)
36999792	Sequencing	CDKN2A as somatic PDAC driver (F010)
28722124	Sequencing	CDKN2A/TP53/SMAD4 PDAC quartet (F010)
40851494	Review	5–10% familial melanoma fraction (F011)

PMID	Role	Supports
32482799	National cohort	172 families, variant spectrum, founder (F011)
17505264	Clinic cohort	CDKN2A rare in unselected melanoma (F011)

Limitations and Knowledge Gaps

1. **Penetrance uncertainty.** Estimates vary several-fold by ascertainment and geography (28% vs 58–91% for melanoma; PDAC largely from one founder cohort). Variant-specific penetrance for most individual CDKN2A alleles is poorly quantified.
2. **p16 vs p14ARF attribution.** Because the locus encodes two proteins, the independent contribution of ARF loss to the human melanoma/PDAC phenotype remains incompletely resolved; most data are variant-mixed.
3. **PDAC surveillance generalizability.** The strongest surveillance evidence comes from the Dutch p16-Leiden cohort; optimal modality, start age, and interval for other variants/populations are less certain, and CAPS cyst criteria are insufficiently sensitive.
4. **Limited direct germline-syndrome therapeutics.** CDK4/6-inhibitor rationale in CDKN2A-null tumors is mechanistically compelling but not validated in randomized trials for this syndrome.
5. **Modifier genetics.** Beyond MC1R, candidate modifiers (e.g., TERT/CLPTM1L) are not established.
6. **No in-house data analysis.** This report synthesizes literature; no primary dataset was analyzed in these iterations.

Proposed Follow-up Experiments / Actions

1. **Variant-specific penetrance registry:** pool international CDKN2A carrier cohorts to produce allele- and modifier-stratified age-specific risk curves for melanoma and PDAC.
2. **Prospective multi-variant pancreatic surveillance trial:** extend Dutch-style MRI/EUS surveillance to non–p16-Leiden variants and diverse populations; incorporate ctDNA/CA19-9 and cyst-fluid biomarkers to improve sensitivity.

3. **ARF-specific functional dissection:** engineer isogenic cell/mouse models separating p16 vs p14ARF loss to quantify each arm's tissue-specific oncogenic contribution.
4. **CDK4/6-inhibitor biomarker trials:** test CDK4/6 inhibitors (± MEK or PI3K/mTOR inhibitors) in CDKN2A-deficient melanoma and PDAC, using the documented CDK4/6 dependence as the biomarker.
5. **GxE quantification:** formal modeling of smoking-cessation benefit on pancreatic/aerodigestive risk in carriers to strengthen counseling guidance.
6. **Comparative/animal-model program:** evaluate naturally occurring CDKN2A-pathway tumors in companion animals (OMIA) for translational models.

Evidence source types: human clinical/cohort (majority), model organism (mouse: F003), in vitro/biophysical (F007), computational (F007), and guideline/consensus (F008). PMIDs cited throughout; direct abstract quotes provided in Key Findings.
